

STEPLESS

*Accessibility shouldn't depend on goodwill.
It should pay the people who build it.*

Decentralized accessibility infrastructure on Arc

THE PROBLEM

"Does that café have a ramp?"

This question has no reliable answer anywhere in the world.

1.3B

people live with a significant disability
— 16% of humanity

<1%

of points of interest worldwide carry
usable accessibility data

0

people are paid to collect it today

The data isn't missing because mapping is hard. It's missing because mapping has always been unpaid labour — done out of goodwill, by people who already have less time and less money than average.

Turn accessibility mapping into paid work

- 1 Contribute**

Photograph a ramp, elevator or accessible restroom. Server-side EXIF GPS is the proof you were there.
- 2 Verify**

A verifier confirms the submission on-chain. No trust in the submitter required.
- 3 Get paid**

USDC lands in the contributor's wallet. Sub-second settlement on Arc.
- 4 Resell**

The dataset becomes an oracle. Consumers pay per query in USDC via x402.

Steps 1-3 take under 30 seconds. Step 4 is why steps 1-3 keep happening after the grant money runs out.

Photo to payment, in under 30 seconds

0s

Photo submitted

EXIF GPS extracted server-side from the image bytes

1s

Registered on Arc

Relayer pays gas in USDC — contributor holds no token

2s

Memo attached

Metadata written natively via Arc's Memo predeploy

~20s

Verified

A verifier confirms the contribution on-chain

~21s

USDC received

Treasury settles to the contributor's wallet

On any other chain this is a promise of payment. Here it is the payment — the contributor is still standing in front of the ramp.

This product is not portable to another chain

USDC-denominated gas

A 0.05 USDC reward is only coherent if gas is also USDC. On a chain with a separate gas token, a contributor must first acquire that token — which is the entire onboarding problem, reintroduced.

Sub-second settlement

The reward has to land while the contributor is still at the location. A payment that arrives tomorrow is a different product with different psychology.

Memo predeploy

Structured metadata (packed lat/lng + photo hash) is attached to the transaction itself with a sequential index — not reconstructed from logs by an off-chain indexer we would have to trust.

x402 + Circle stack

Metered, per-query USDC settlement is what turns a grant-funded dataset into a business.

Architecture

- 1 User (browser / Android)**
Photo + GPS. Holds no gas token.
- 2 /api/upload**
Extracts EXIF from the image BYTES. dataHash = keccak256(bytes). Client declares nothing.
- 3 /api/relay**
Serverless relayer (EOA). Validates photo GPS against the declared location. Pays gas in USDC.
- 4 Memo predeploy 0x5294E9...e505**
Relayer calls Memo with the oracle as target; callFrom preserves msg.sender. Native metadata + sequential index.
- 5 SteplessOracle → RewardDistributor**
Location registry, on-chain verification, USDC treasury settlement.
- 6 X402API**
Metered oracle reads. HTTP 402, settled in USDC.

Who pays for the data

Every "earn crypto for contributing data" project dies in the same place: the treasury empties. So half of Stepless is demand, not supply.



Routing & travel apps

Wheelchair-accessible route planning needs ground truth that no map provider sells.



Municipalities

Auditing accessibility compliance and public works against independent field data.



Delivery & mobility platforms

Entrance and step data changes drop-off feasibility and courier routing.

X402API.sol

GET /oracle/locations

→ 402 Payment Required

→ pay in USDC

→ 200 OK + data

Per-query settlement in USDC. Contributor rewards become funded revenue instead of a subsidy with an end date.

One product, both tracks

DeFi

- USDC treasury with on-chain, conditional release — payout is gated by an independent verification, not a signature
- Multi-step settlement: register → verify → distribute, each step on-chain
- Metered revenue settled in USDC via x402
- Relay treasury management with failure recovery for reverted rewards

Agentic Economy

- The oracle's consumers are agents: a routing agent queries, gets a 402, pays USDC, proceeds — no human on either side
- The relayer is an autonomous spender: it decides, funds gas and settles rewards against on-chain signals
- Decision logic tied to real signals — EXIF GPS, place existence checks, contributor reputation, risk scoring
- Agent-to-agent payment for a data service, priced per call

What is live right now — and what is not

Arc Testnet • chain 5042002 • verified against ArcScan on 9 Aug 2026

SteplessOracle	0x69b3f9ca...a5cc	Live • last tx 7 Aug
RewardDistributor	0xef5d148b...2ea0	Live • 24.6 USDC treasury
X402API	0x0D318864...fCc9	Live • settling in USDC

Also live: the app, the Android build, the serverless relayer, EXIF anti-fraud, place-existence checks and contributor reputation scoring.

Not live: mainnet.

Arc mainnet launches 16 Sep 2026 and Circle has not published mainnet USDC and Memo addresses yet. The arc-mainnet entry in config/networks.json carries null fields on purpose and the backend refuses to boot against it. On Arc, a call to an address with no code returns success — a guessed address would have the contract mark rewards as paid without moving a cent. We would rather not ship than ship that.

Where this goes

Now

Testnet MVP

Contracts live, app shipped, treasury paying, Android build published.

Sep 2026

Mainnet

Deploy the day Circle publishes mainnet addresses. Contract verification and subgraph indexing land with it.

Q4 2026

First paying consumer

One routing app or municipality on x402. This is the metric that matters — not contributor count.

2027

Protocol

Open verifier set with staking, dispute resolution, and a contributor DAO over reward parameters.

What we want from the accelerator

Introductions on the demand side. The engineering is the part we can do alone; a signed pilot with one mobility platform or one city is what turns this from a hackathon project into a protocol.

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stepless.vercel.app · github.com/Acarlosr · Arc Testnet 5042002